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Paleomagnetic evidence for tectonic rotation of the Oregon Coast Range

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ABSTRACT

Paleomagnetic directions from lower, middle, and upper Eocene sedimentary and volcanic rocks in the Oregon Coast Range point 50° to 70° east of the expected Eocene field direction. This discrepancy is attributed to the tectonic rotation of a block at least 225 km in length extending from just north of the Klamath Mountains to north of Newport, Oregon. Two models are developed for the tectonic evolution of the Pacific Northwest. In the first, the coastal block is rotated seaward about a pivot point near its southern end in order to restore it to its Eocene position. In the second, it is assumed that the coastal block extends all the way to the Olympic Peninsula. Rotating the block back about a pivot near its northern end moves the southern end and the Klamath Mountains eastward across Oregon and restores the Klamath metamorphic belts to a position against the line of Cretaceous batholiths in northwestern Nevada.

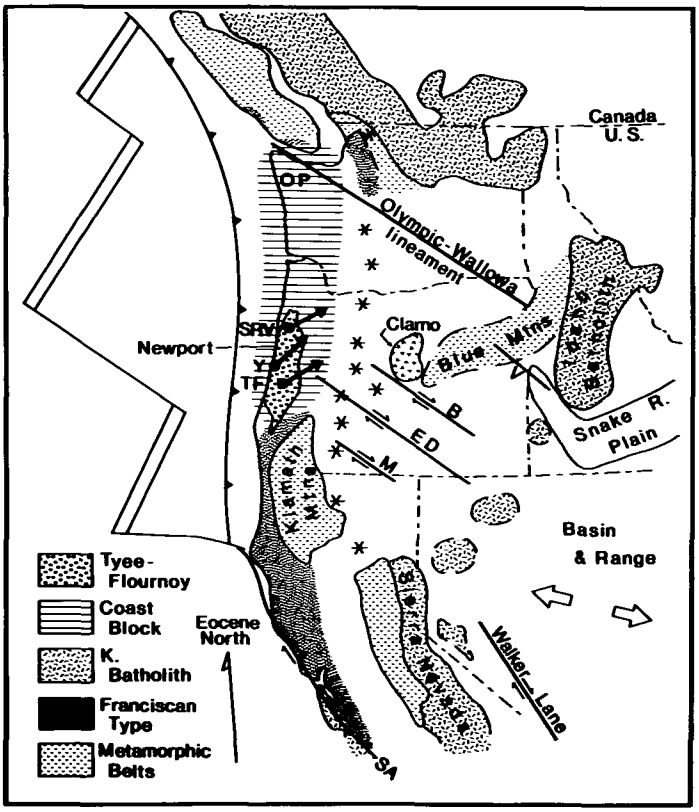


Figure 1. Generalized geologic and tectonic map of northwestern United States based in part on Hamilton (1969) and Lawrence (1976). SRV = Siletz River Volcanic Series, Y = Yachats basalt, TF = Tyee and Flournoy Formations, OP = Olympic Peninsula. Fault zones and lineaments: V = Vale, B = Brothers, ED = Eugene-Denio, M = McLoughlin, SA = San Andreas.

INTRODUCTION

We have new paleomagnetic data from the Oregon Coast Range that confirm and extend the anomalous magnetization directions discovered by Cox (1957, 1959) for the Siletz River Volcanic Series. Cox found a paleomagnetic direction for these lower and middle Eocene basalts that pointed 70° east of north—a clockwise rotation of at least 75° from the expected Eocene field direction (Fig. 1).

The Siletz River Volcanic Series is correlative with other submarine and subaerial basalts including volcanics of the Roseburg Formation in the southern Oregon Coast Range, the Crescent Formation in the Olympic Peninsula, and the Metchosin Volcanics on Vancouver Island (Snavely and others, 1968; Baldwin, 1974; Cady, 1975). These formations comprise the oldest rocks exposed in the Coast Ranges of Oregon and Washington. They have been interpreted as oceanic-ridge tholeiites overlain by tholeiitic and alkalic basalts forming seamounts and islands petrologically similar to the Hawaiian Islands (Snavely and others, 1968; Glassley, 1974; Cady, 1975; for an alternate interpretation see Lyttle and Clarke, 1975, 1976).

Available geophysical evidence in the form of seismic refraction lines (Tatel and Tuve, 1955; Berg and others, 1966) and velocity structure inferred from earthquakes in the Puget Sound area (Langston and Blum, 1977) indicates a crustal thickness of only 15 to 20 km under the Coast Ranges. This thickness is consistent with that expected for a seamount province or an oceanic aseismic ridge that has become accreted to the margin of the continent (Snavely and MacLeod, 1974; Dickinson, 1976).

One could argue that the rotated paleomagnetic results are not surprising in this tectonic setting—especially given the obvious rotation of sea-floor magnetic stripes observed in small plates caught between larger plates (Silver, 1971) and given the consistent

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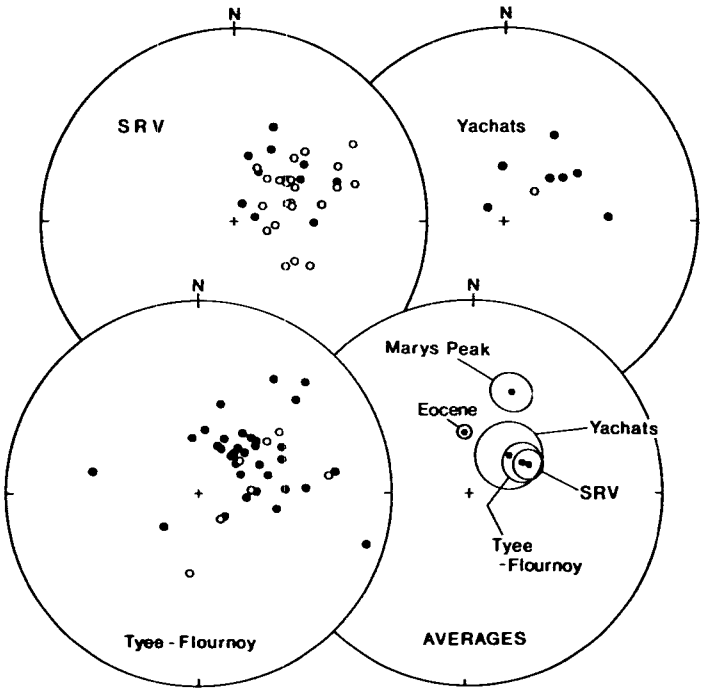


Figure 2. Site-mean magnetization directions for Siletz River Volcanic Series (SRV), Yachats basalt, Tyee and Flournoy Formations, and averages. Eocene field direction predicted from studies in stable parts of North America and result for Marys Peak sill (Clark, 1969) are also shown. Reversed polarities have been projected through origin onto lower hemisphere and are shown as open circles. Large ovals are circles of 95% confidence about the means.

TABLE 1. PALEOMAGNETIC DATA

	D (°)	I (°)	Field α_{95}	No. sites	Pole lat (°)	Pole long (°E)	Pole α_{95}	Rotation*
Expected field directions for Eocene rocks (from North American studies, see text)	355	64	4	4	87	169	6	. .
Expected field directions for Eocene rocks (from North American, European, and Russian studies, see text)	345	68	3	19	78	187	5	. .
Expected field directions for upper Tertiary rocks (Beck, 1976)	354	61	4	3	85	114	6	. .
Siletz River Volcanic Series (Cox, 1957)	70	55	7	8	37	310	9	75 ± 12
Siletz River Volcanic Series (Cox, 1959, recalculated)	68	56	7	9	39	310	9	73 ± 12
Siletz River Volcanic Series (present study)	60	62	8	24	48	306	11	65 ± 17
Siletz River Volcanic Series (Cox, 1959, recalculated, and present study)	63	61	6	33	45	307	8	68 ± 12
Tyee-Flournoy Formations	59	63	8	40	49	305	11	64 ± 16
Yachats basalt	46	64	14	8	58	308	20	51 ± 33
Marys Peak sill (Clark, 1969)	22	42	8	26	63	8	8	28 ± 12
Miocene basalts from Oregon Coast	354	52	25	8	67	349	29	0 ± 44

Note: D = declination, I = inclination.
* ± indicates 95% confidence limits.

clockwise rotation of magnetization directions found in many paleomagnetic studies of rocks along the west coast of North America (Beck, 1976). Beck has attributed these rotations to a style of "ball bearing" tectonics for crustal blocks caught in a broad shear zone between northward-moving oceanic plates and the North American plate.

Our new data indicate that this ball bearing—if it can be regarded as such—extends at least 225 km from just north of the Klamath Mountains to beyond Newport, Oregon. Significant rotation is recorded not only in the lower Eocene Siletz River Volcanic Series but also in the middle Eocene marine sediments of the Tyee and Flournoy Formations that blanket this area (Snavely and others, 1964; Lovell, 1969; Baldwin, 1974) and in the upper Eocene volcanics of the Yachats basalt south of Newport (Snavely and MacLeod, 1974). The apparently "rigid" behavior of this block since middle Eocene time is also attested to by the remarkable consistency of south to north current directions recorded in the middle Eocene turbidites of the Tyee and Flournoy Formations (Snavely and others, 1964).

Because the middle Eocene marine section appears to be derived in large part from the Klamath Mountains and to lie in depositional contact upon the metamorphic rocks of the Klamaths, a tectonic rotation of the Coast Range would appear to have tectonic implications for the Klamaths also. Such a relation has been implied by various authors who have cited Cox's anomalous paleomagnetic vector for the Siletz River Volcanic Series in support of Carey's (1958) orocline or Wise's (1963) megashear. These hypotheses attempt to explain the bend in the Cordillera expressed by the anomalous position of the Klamaths and the discordant northeast trend of the Blue Mountain metamorphic belts in eastern Oregon as secondary strains imposed upon an originally linear orogenic belt (Carey, 1958; Wise, 1963; Irving, 1964; Hamilton and Myers, 1966; Hamilton, 1969; Watkins and Baksi, 1974). In contrast with the simple ball-bearing model, these theories suggest that the rotation of the Coast Range has broad implications for the tectonics of the Pacific Northwest.

PALEOMAGNETIC DATA

Our new results are from 24 additional flows and sills of the Siletz River Volcanic Series, from 40 sites in the middle Eocene marine section between the Klamath Mountains and Newport, and from 8 flows of the upper Eocene Yachats basalt south of Newport. The geologic history of the sampling area has been summarized by Snavely and others (1969), and the units sampled are described in detail in references cited above. Most of our samples from the middle Eocene marine section come from the Tyee and Flournoy Formations, with a few sites in the Lookingglass and Elkton formations of Baldwin (1974). The paleomagnetic data will be discussed at greater length in a future publication.

All samples have been magnetically cleaned using an alternating field demagnetizer at peak fields ranging from 100 to 320 oersteds. Outcrops with circles of 95% confidence having radii greater than 25° have been discarded. The paleomagnetic vectors (Fig. 2; Table 1) are consistently rotated 50° to 75° clockwise from the expected Eocene geocentric dipole direction determined for tectonically stable areas (Fig. 3). The Marys Peak sill east of Newport, sampled by Clark (1969) and dated at 29.9 m.y. (Parke Snavely, personal commun.), shows a lesser degree of rotation (Table 1).

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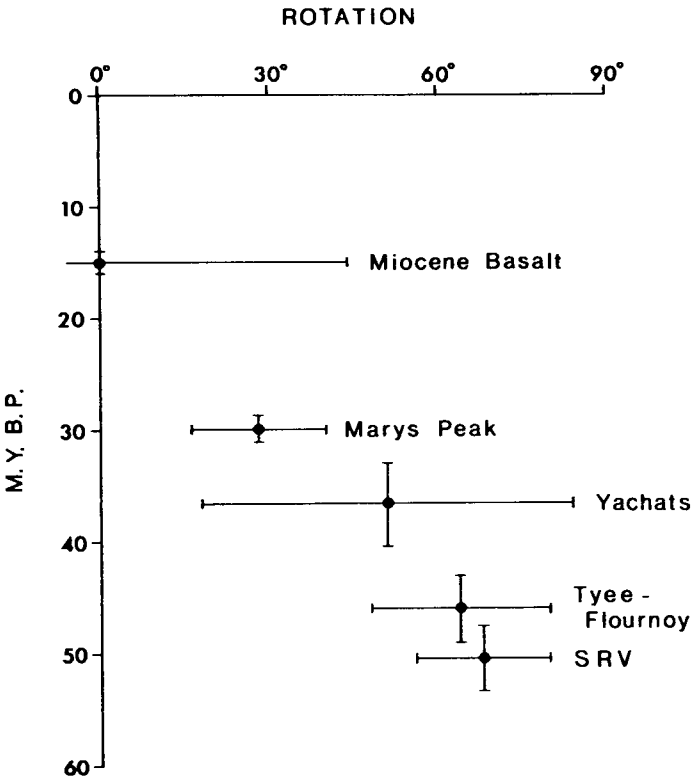


Figure 3. Rotation versus age for units in Oregon Coast Range, using time scale of Berggren and Van Couvering (1974).

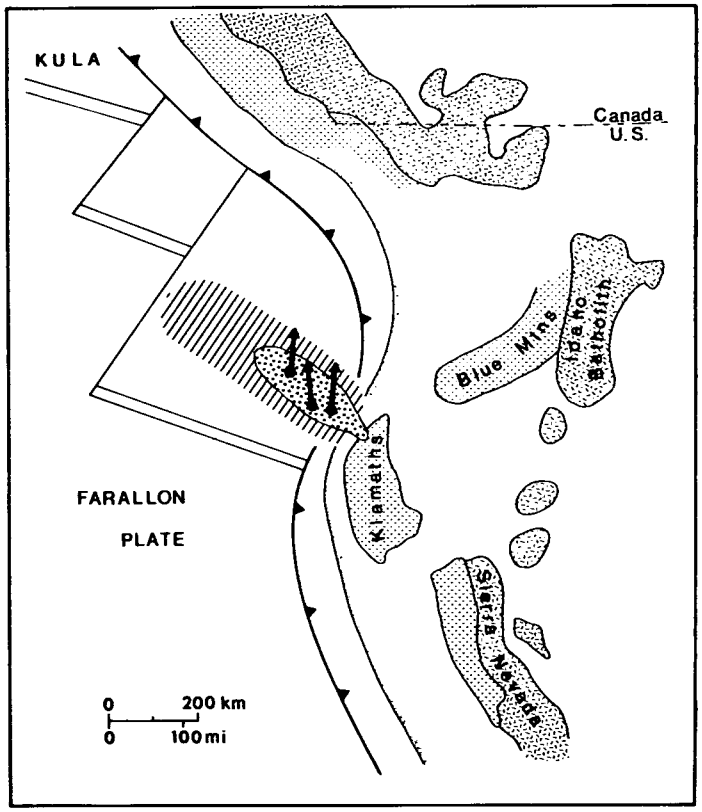


Figure 4. First model: Pacific Northwest in middle Eocene time, showing Coast Range block as part of small plate related to northward migration of Kula-Farallon-North America triple junction. Compare Silver (1971) for possible present-day analog. Basin and Range province has been contracted about 180 km at latitude of northern Sierra Nevada. Patterns as in Figure 1.

Only four paleomagnetic studies have been made of Eocene rocks from stable portions of North America (Currie and Larochelle, 1969; Lovlie and Opdyke, 1974; Shive and Pruss, 1977; Shive and others, 1977). These give an average pole somewhat different from that obtained by using poles from Africa, Europe, and Russia (Irving and others, 1976) and closing the Atlantic back to its Eocene position using the spreading history of Pitman and Talwani (1972). For either set of poles, however, the expected field direction in the Pacific Northwest is at least 5° to the west of north (Table 1). We feel that it is extremely unlikely that our rotated magnetization directions could be recording an excursion or a local anomaly of the geomagnetic field because we have sampled both normally and reversely magnetized rocks and because the ages of the rocks span at least 15 m.y.

INTERPRETATION

We propose two possible paleogeographic reconstructions that restore the paleomagnetic vectors from the Coast Range to the expected Eocene direction.

The first rotates the Coast Range block about a pivot near its southern end and swings the northern end seaward (Fig. 4). In this restoration the middle Eocene sediments were deposited on a slab of oceanic plate that was subsequently rotated during subduction. There are several problems with this model. Although the lower Eocene basalts in the south show signs of being subducted under the Klamaths (Baldwin, 1974), the rotated middle Eocene marine section is only very gently deformed with beds generally tilted less than 10° to 15°, and the rotated upper Eocene Yachats basalt was extruded in a tectonic environment thought to have extensional stress axes oriented perpendicular to the present coast line (Snavely and MacLeod, 1974). Also, continuity of geologic trends and rock types (Parke Snavely, personal commun.) suggest that the rotated coastal block extends all the way from the Klamath Mountains to the Olympic Peninsula. This presents a difficulty in that the continent-derived turbidites and shales interbedded with the basalts in the Crescent Formation on the Olympic Peninsula, including quartz diorite boulders 3 m in diameter (Cady, 1975), seem to require that the northern end of the block remain close to the continental margin as the basalts were extruded. An undetected suture may exist somewhere in the Coast Ranges between the Olympic Peninsula and Newport so that the southern portion could have been rotated seaward while the Crescent Formation stayed near the continent. Additional paleomagnetic work in the Coast Ranges should clarify this possibility.

A second model fits the ideas of Hamilton (1969) for rifting the Klamath Mountain metamorphic belts away from the line of Cretaceous batholiths in northwestern Nevada and keeps the Olympic Peninsula near the continental margin (Fig. 5). In this model, the Coast Range block is rotated about a pivot near its northern end. To restore the paleomagnetic vectors, the southern end of the Coast Range block and the Klamath Mountains must be swung back across most of Oregon until the block lies roughly

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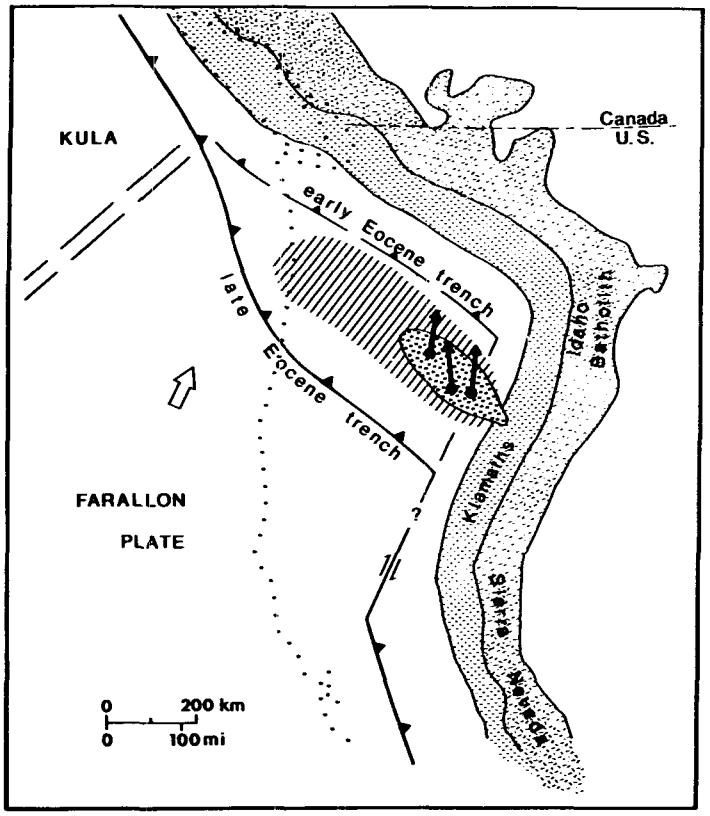


Figure 5. Second model: Pacific Northwest in middle Eocene time, showing Coast Range block accreted to edge of continent and about to start rotating. Subsequent rifting of metamorphic belts is similar to that envisioned by Hamilton (1969). Patterns as in Figure 1.

parallel to the Olympic-Wallowa lineament. Skehan (1965a, 1965b) suggested that this line forms a boundary between continental crust to the north and crust of a more oceanic type to the south. A poorly defined early Tertiary andesitic magmatic arc also runs roughly parallel to this lineament across northern Washington, Idaho, and into Wyoming (Dickinson, 1976; Snyder and others, 1976).

In this second model, an early Tertiary subduction zone trended parallel to this lineament. An oceanic aseismic ridge, possibly generated by the hotspot now under Yellowstone, clogged this old zone, causing it to jump seaward in the middle Eocene. The middle Eocene marine section in the Oregon Coast Range was deposited from a Klamath source northwestward into a fore-arc basin behind the new trench. In late Eocene time, the south end of the present Coast Range block and the Klamath Mountains began rotating seaward because of back-arc spreading behind the new trench, with rifting perhaps aided by the presence of the hotspot. Changes in the stresses acting on the coastal block could have contributed to the rifting: about 40 m.y. ago the Pacific plate turned from a northward to a more westerly direction (Jarrard and Clague, 1977), and the absolute motion of the North American plate over the mantle appears to have decreased (Coney, 1972). Also, between 60 and 40 m.y. ago, the ridge separating the Kula and Farallon plates moved past the coasts of Oregon and Washington (Atwater, 1970). All of these could have changed the stresses confining the coastal block, permitting it to swing out to the west.

All but about 28° of the rotation occurred by the time of emplacement of the Marys Peak sill 30 m.y. ago. The Blue Mountains in eastern Oregon were rifted away from the Idaho batholith during the rotation of the Coast Range. The gaps created by this rifting might be filled either by attenuation of the old crust or by extrusion of Eocene volcanics which, if they exist, are mostly hidden by younger volcanics in eastern Washington and Oregon.

This second model runs into at least two problems. One is that the andesites of the Clarno Formation in central Oregon, which have a radiometric age as old as 41.0 m.y. (Swanson and Robinson, 1968), appear from limited paleomagnetic evidence (Whitney, 1974; Sherman Grommé, personal commun.) to be rotated by no more than 15°. This requires either very rapid rotation of the coastal block or westward rafting of the Clarno Formation behind the coastal block without significant rotation. The other problem is that studies of the Klamath Mountains in California have led Blake and Jones (1977), Irwin (1977), and Schweikert (1976) to postulate that the offset of the Klamaths from the metamorphic belts of the northern Sierra Nevada occurred by rifting sometime in Mesozoic time rather than in Tertiary time. Additional paleomagnetic work in central Oregon and in the Klamaths is needed to determine if an orocline does exist and to pin down its time of formation.

The last 28° of rotation may be related to the extension of the Basin and Range province. Watkins and Baksi (1974) have noted paleomagnetic evidence for small amounts of oroclinal bending within the Miocene Columbia River Basalt. Lawrence (1976) interprets the west-northwest-trending fault zones and lineations that cross Oregon as active strike-slip faults separating zones characterized by normal block faulting with successively greater amounts of extension in the zones to the south. (The Mitchell and John Day faults in central Oregon may be earlier analogs of Lawrence's strike-slip zones accommodating greater amounts of rifting to the south.) The apparent failure of Lawrence's west-northwest-trending zones to cut the Coast Range suggests that the coastal block may still be rotating today in response to differential extension behind it.

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